

3.2.P.5.1 Specifications

The specifications for the mRNA-1273.251 Drug Product are provided in [Table 1](#). It applies to the 3.2 mL fill volume multiple-dose vial (MDV) presentation.

Table 1: mRNA-1273.251 Drug Product – 0.10 mg/mL - MDV

Test	Method	Acceptance Criteria	
Appearance	Visual, USP<631>, USP<790>, Ph. Eur., 2.2.2, Ph. Eur. 2.9.20, JP 6.06	White to off-white dispersion.	
		May contain visible white or translucent product-related particulates.	
Total RNA content	Anion Exchange (AEX) HPLC (SOP-0999)	4.2 1st ind	
RNA Identity ¹	Reverse Transcription/Sanger (SOP-1337)	Sequence Matches Description	
Purity ²	RP-IP-HPLC (SOP-1142)	4.2 1st ind	
Product-related impurities ²			
% RNA Encapsulation			Absorbance (SOP-1000)
In Vitro Translation			Methionine Labelling (SOP-0937)
Lipid identification			
SM-102	HPLC-CAD (SOP-1001)	Matches retention time of reference	
Cholesterol		Matches retention time of reference	
DSPC		Matches retention time of reference	
PEG2000-DMG		Matches retention time of reference	

Test	Method	Acceptance Criteria
Lipid content		
SM-102	HPLC-CAD (SOP-1001)	4.2 1st ind
Cholesterol		
DSPC		
PEG2000-DMG		
Lipid impurities		
Particle size	Dynamic Light Scattering (SOP-0998)	4.2 1st ind
Polydispersity		
pH	USP <791>, Ph. Eur. 2.2.3	
Osmolality	USP <785> Freezing Point Depression Ph. Eur. 2.2.35	
Particulate matter		
≥ 25 μm	USP <788> (Method 2), Ph. Eur. 2.9.19, JP 6.07	4.2 1st ind
≥ 10 μm		
Container content	USP <697>, Ph. Eur. 2.9.17	4.2 1st ind
Bacterial endotoxin	USP <85>, Ph. Eur. 2.6.14, JP 4.01	
Sterility	USP <71>, Ph. Eur. 2.6.1, JP 4.06	No growth
Container Closure Integrity (Stability only)	Helium Leak Detection (803-TM, 895 TM)	Passes

Abbreviations: RRT = relative retention time.

¹ To be tested on Unlabelled Drug Product (UDP) and Labelled Drug Product LDP (test and report ID)

² To be executed on the LDP only.

³ The collection of samples is taken at LDP stage.